

Utkarsh Pathak

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PROFESSIONAL SUMMARY

Speech researcher and engineer with specialized experience in multilingual Text-to-Speech systems, zero-shot synthesis, phoneme modeling, and production deployment. Led development of 22 official-language TTS systems deployed at scale for educational translation. Research published at INTERSPEECH 2025.

EDUCATION

M.S. (Research), Data Science and Artificial Intelligence

2023 – Present

under Prof. Hema A. Murthy/ Prof. Mitesh Khapra

Indian Institute of Technology Madras

CGPA: 8.02/10

B.E., Computer Science and Engineering

2016 – 2020

Chameli Devi School of Engineering

CGPA: 8.33/10

RESEARCH EXPERIENCE/PUBLICATIONS

Kinship in Speech: Leveraging Linguistic Relatedness for Zero-Shot TTS in Indian Languages

Accepted at INTERSPEECH 2025, Netherlands

- Proposed a zero-shot TTS framework leveraging cross-lingual phoneme alignment via an extended Common Label Set (CLS) of phonemes.
- Designed phonotactic-aware text parsing and synthesizer selection enabling intelligible synthesis for unseen languages without parallel data.
- Demonstrated cross-family generalization (e.g., Sanskrit) and dialect-aware synthesis across language families (Maharashtrian and Canara Konkani).
- Conducted subjective and objective evaluations validating robustness across scripts and phonotactic constraints.

Living Among Languages: A Framework for Code-Switching TTS Across Indian Languages

Submitted to INTERSPEECH 2026, Sydney

- Proposed an architecture-agnostic framework for code switching TTS across Indian languages and English.
- Improved and extended the CLS for 22 official Indian languages.
- Demonstrated how seamless code switching/mixing and multilingualism can be achieved while retaining the native accent and without retraining.

PROFESSIONAL EXPERIENCE

Project Officer – Speech Technologies in Indian Languages

2023 – Present

Indian Institute of Technology Madras

- Led training and optimization of 22 bilingual TTS models using FastSpeech2 and HiFi-GAN across official Indian languages.
- Designed multi-speaker and multilingual training strategies for improved robustness and generalization.
- Optimized inference pipelines for CPU deployment, enabling scalable low-latency synthesis.
- Collaborated with other IITs for building and deploying an end-to-end educational translation pipeline: ASR → MT → TTS → Lip-sync video generation.
- Developed Transformer-based (T5) grapheme-to-phoneme model for Hindi, improving phoneme prediction.
- Open-sourced multiple models and components; deployed systems on NPTEL servers and annotation platforms.

Senior Systems Engineer – Microsoft D365 CRM

2021 – 2023

Infosys Limited, Pune

- Developed and maintained enterprise-grade cloud services using .NET, C#, Angular, and Microsoft Azure.
- Managed production deployments, debugging, and performance optimization in live environments.
- Strengthened end-to-end system thinking: architecture, reliability, and scalability.

SELECTED TECHNICAL PROJECTS

Domain Term Discovery for Machine Translation

- Improved transcription accuracy for extempore educational lectures using LLM-based post-editing.
- Curated large-scale dataset using GPT-3.5 Turbo for domain-term identification.
- Trained BERT-based classifiers and experimented with few-shot learning for terminology preservation.

Transformer-Based Grapheme-to-Phoneme Modeling

- Implemented T5-based seq2seq model for Hindi phoneme prediction.
- Improved phoneme accuracy and reduced manual intervention in multilingual TTS pipelines.

SELECTED COURSE PROJECTS

Deep Learning – Neural Networks and Sequence Modeling

IIT Madras

(Prof. Mitesh Khapra)

- Implemented neural networks and backpropagation from scratch; developed optimization algorithms including SGD, Momentum, NAG, RMSProp, Adam, and Adagrad.
- Built CNN-based image classifier and performed feature visualization and fine-tuning of pre-trained models.
- Conducted comparative analysis of RNN, LSTM, and GRU architectures for machine transliteration; improved performance using attention mechanisms (used WANDB).

Pattern Recognition and Machine Learning

IIT Madras

(Prof. Arun Rajkumar)

- Implemented PCA and kernel-PCA for dimensionality reduction; developed clustering models using GMM optimized via EM algorithm.
- Implemented Naive Bayes and AdaBoost from scratch for classification tasks.

Natural Language Processing

IIT Madras

(Prof. Sutanu Chakraborti)

- Implemented language modeling, POS tagging, syntactic parsing, and a retrieval-based information retrieval system.

RELEVANT COURSEWORK

Pattern Recognition and Machine Learning (CS5691), Deep learning (CS6910), Natural Language Processing (CS6370), Linear Algebra, Random Probabilities and Optimisation(DA5000), Advanced Topics in Signal Processing (EE7130).

TECHNICAL SKILLS

Core Areas: Multilingual TTS, Zero-Shot Synthesis, Cross-Lingual Phoneme Modeling

Deep Learning: PyTorch, ESPnet, HuggingFace, Scikit-learn

Programming: Python, C++, Java

Deployment: CPU Optimization, Git